Edison Wang

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Education

University of Waterloo Sept 2025 - May 2030
Bachelor of Computer Science *GPA: 3.7/4.0*
Relevant Coursework: Object Oriented Programming (Enriched), Logic and Computation

Technical Skills

Languages: Python, TypeScript, Go, JavaScript, SQL, C, HTML/CSS
Frameworks/Libraries: FastAPI, Next.js, Django REST Framework, React, Tailwind CSS
Tools/Technologies: Git, GitHub, Node.js, Docker, Kubernetes, MS Azure, Linux, VS Code, Claude Code
Databases: PostgreSQL, MongoDB, Redis, MS SQL Server

Work Experience

Software Developer Intern May 2026 - Aug 2026
Government of Ontario (MPBSDP) *North York, ON*

- Developed a modular regression testing framework for an ASP.NET WebForms healthcare portal supporting over **5k daily users**, leveraging TypeScript, Playwright and Node.js to achieve **98% critical feature coverage**
- Implemented adaptable CI pipeline with Azure DevOps to enforce code quality and application functionality, integrating Azure OpenAI API via reporter hook to diagnose pipeline failures **60% faster** than manual debugging
- Authored SQL scripts and stored procedures to sanitize **100+ tables** across MS SQL Server databases, cutting data cleanup times by **95%** versus manual workflows and automating execution via daily asynchronous jobs

Software Lead (Prev. PM, Backend Developer) Sept 2025 - Present
UW Orbital *Waterloo, ON*

- Led **10-member** fullstack subteam in building ground station software, securing **1st** at CSDC-7 among **8 teams**
- Implemented OAuth2/OIDC authentication and RBAC via self-hosted Keycloak on Linux to secure satellite communication infrastructure, supervised with Runit for **98% uptime** and load-tested to handle **6000+ RPM**
- Migrated frontend and backend applications out of a **393k-line** firmware repository, isolating **45k+ lines** of application code, cutting CI build time by **60%**, and preserving cross-system integration via Git submodules
- Developed mission-critical Data Access Layer in FastAPI to reduce SQLModel ORM code duplication by **50%**
- Engineered automated ETL pipeline and internal CLI tool using Python scripts, processing and migrating **100k+ rows** of government data to validate amateur radio operator certifications for public-facing usage

Projects

Resumakr (LaTeX Compiler Microservice) | *Go, gRPC, Docker, GitHub Actions, MS Azure, LaTeX*

- Built a LaTeX processing gRPC server in Golang, compiling resumes at **150+ files/sec** on Azure Container Apps
- Improved image build process via Dockerfile optimization to achieve **<100MB** image size and **83% faster** builds
- Enhanced request throughput by precompiling LaTeX dependencies, resulting in **50ms PDF compilations**

YYCirrus (Local Kubernetes Service) | *Kubernetes (K3s), ArgoCD, Ubuntu Linux, Tailscale, Cloudflare*

- Architected a multi-node Kubernetes cluster homelab using Windows machines with WSL, providing **99.5% uptime per month** for **10+ services**, including databases, production APIs, and auxiliary workloads
- Designed CI/CD pipelines with GitHub Actions, ArgoCD, and GitOps repo, automating **50+ app deployments**

Memento (AI-powered Memory System) | *Python, FastAPI, MCP, Typescript, React, MongoDB, GitHub Apps*

- Built a fullstack developer tool using Backboard.io API, combining documentation files, Claude Code sessions, PR diffs, and GitHub repositories to generate grounded institutional memory, accessible by **10 MCP server tools**
- Implemented memory distillation pipeline with atomic job claiming, automatic retry, and crash recovery, utilizing a consistency-checking system to alert contradictions between stored memories and evolving code in **<10 seconds**
- Created interactive knowledge graph via react-force-graph-2d to visualize **100+** decisions in **5k-line** mock project

Professional Certificates

Meta Back-End Developer Certificate July 2025 - Aug 2025
Meta *Remote*